Course Code : MAT 255 **GVHW/RW – 23 / 1431**

Third Semester B. Tech. (Electronics and Communication Engineering) Examination

ENGINEERING MATHEMATICS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

Use of Non – programmable calculator is permitted.

1. Solve any **Two** :

(a) Determine whether the following functions are analytic or not :

(i) $z^3 + \dfrac{1+i}{2z}$,

(ii) $\dfrac{z}{x^2 + y^2}$,

where $z = x + iy$. 6(CO2)

(b) Given that $f(z) = u + iv$ is an analytic function and $u + v = e^x(\cos y + \sin y)$. Find $f(z)$. 6(CO2)

(c) Find the Mobius transformation which maps the points $z_1 = 2$, $z_2 = i$, $z_3 = -2$ onto the points $w_1 = 1$, $w_2 = i$ and $w_3 = -1$. 6(CO2)

2. Solve any **Two** :

(a) Find the singular points and residues for the function :

$$\frac{z+2}{(z-2)\,(z+1)^2}$$

6(CO2)

(b) Evaluate by contour integration : $\int_0^\infty \dfrac{dx}{a^4 - x^4}$, $a > 0$. 6(CO2)

(c) Find the Laurent's expansion for $f(z) = \dfrac{z^2 - 1}{z^2 + 5z + 6}$ in the region given by $2 < |z| < 3$. 6(CO2)

3. Solve any **Two** :

(a) Solve the following Lagrange's linear equation :

$2p + 3q = \sin(x) + \sin(y)$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. 6(CO1)

(b) Solve : $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 2\frac{\partial^2 z}{\partial y^2} = (y-1)e^x$. 6(CO1)

(c) Find the solution of telegraph equation $\frac{\partial^2 v}{\partial x^2} = RC\frac{\partial v}{\partial t}$ by separation of variable method. 6(CO1)

4. Solve :

(a) Find Laplace Transform of following :

(i) $\cos^2 3t + \sin 5t \sin 2t$,

(ii) $\frac{1 - e^{-at}}{t}$.

6(CO3)

(b) Solve by Laplace Transform method :

$$\frac{d^2 y}{dt^2} + 9y = 18t,$$

given that $y(0) = 0$ and $y(\pi/2) = 0$. 6(CO3)

5. Solve :

(a) Find Z-transform of following :

(i) $Z\{n+2\}$,

(ii) $Z\{3 \cdot 2^n + 4 \cdot (-1)^n\}$,

(iii) $Z\{a^n \cos nt\}$. 6(CO3)

(b) Solve by Z-transform method : $y_{n+2} - y_n = 2^n$ where $y_0 = 0$, $y_1 = 1$.
6(CO3)